Sri Lanka Institute of Information Technology



Lab Submission
<Lab sheet 06>

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Discrete Mathematics | IT1160

B.Sc. (Hons) in Information Technology

Part 01

In [1]:
```
def S(n):
    if n == 0:
        return 0
    else:
        return n + S(n - 1)

print("S(5) =", S(5))
```

```
S(5) = 15
```

In [3]:
```
def L(n):
    if n == 0:
        return 10
    else:
        return L(n - 1) + 5

print(L(1))
print(L(2))
print(L(3))
```

```
15
20
25
```

In [4]:
```
def harmonic_sum(n):
    if n == 1:
        return 1
    else:
        return 1/n + harmonic_sum(n - 1)

print("harmonic_sum(5) =", round(harmonic_sum(5), 4))
```

```
harmonic_sum(5) = 2.2833
```

In [5]:
```
def bacteria(n):
    if n == 1:
        return 2
    elif n == 2:
        return 3
    else:
        return bacteria(n - 1) * bacteria(n - 2)

print("Day 1 =", bacteria(1))
print("Day 2 =", bacteria(2))
print("Day 3 =", bacteria(3))
print("Day 4 =", bacteria(4))
```

```
Day 1 = 2
Day 2 = 3
Day 3 = 6
Day 4 = 18
```

Part 02

In [6]:
```
def pascal(n, k):
    if k == 0 or k == n:
        return 1
    else:
        return pascal(n - 1, k - 1) + pascal(n - 1, k)

print(pascal(4, 2))
```

```
6
```

In [7]:
```
def sum_of_digits(n):
    if n == 0:
        return 0
    else:
        return n % 10 + sum_of_digits(n // 10)

print("sum_of_digits(1234) =", sum_of_digits(1234))
```

```
sum_of_digits(1234) = 10
```

In [8]:
```
def gcd(a, b):
    if b == 0:
        return a
    else:
        return gcd(b, a % b)

print("gcd(48, 18) =", gcd(48, 18))
```

```
gcd(48, 18) = 6
```

In [9]:
```
def sum_even(n):
    if n < 2:
        return 0
    elif n % 2 != 0:
        return sum_even(n - 1)
    else:
        return n + sum_even(n - 2)

print("sum_even(8) =", sum_even(8))
```

```
sum_even(8) = 20
```

In [10]:
```
def savings(n):
    if n == 1:
        return 1000
    elif n == 2:
        return 1050
    else:
        return 1.05 * savings(n - 1) + savings(n - 2)

print("Month 1 =", savings(1))
print("Month 2 =", savings(2))
print("Month 3 =", savings(3))
print("Month 4 =", savings(4))
```

```
Month 1 = 1000
Month 2 = 1050
Month 3 = 2102.5
Month 4 = 3257.625
```

In [11]:
```
def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n - 1)

def sum_factorials(n):
    if n == 0:
        return 0
    else:
        return factorial(n) + sum_factorials(n - 1)

print("sum_factorials(4) =", sum_factorials(4))
```

```
sum_factorials(4) = 33
```